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B Ravi Kiran

R&D Engineer, AI-ML for ADAS
AKKA Tech. Autonomous Systems.

Overview

- **Industry**
  - **AI & ML R&D Engineer** Autonomous systems team,AKKA Technologies, Guyancout.
  - **Vision & Deep learning Consultant** at Uncanny vision, Paris(remote).
  - **Software design engineer** Experience in ARM/DSP embedded environment, Texas Instruments, India.
  - **Strong programming skills** in C++, Python, C, Matlab.
- **Academic**
  - **ATER** (Temporary professor) Université Lille 3 and CRIStAL lab, INRIA-Lille Nord.
  - **PostDoc II** ENS Paris & ThalesAlenia Space, Time series anomaly detection.
  - **PostDoc I** CAOR, Mines ParisTech, Hyperspectral image classification for tumor detection.
  - **PhD** Computer Science, Universite Paris-Est (LIGM-ESIEE). **Best thesis** in Mathematics & STIC
  - **20 publications** in vision & ML, in leading confs (ECCV), and journals (Pattern recognition).
- **Focus** Perception (Vision/Lidar), ML, Deep learning, Deep reinforcement learning.

Skills

Programming	Python, C++, C, Matlab, Git/SVN, Latex/Beamer.
Packages	Pandas, Scikit-learn, Scikit-image, OpenCV, Tensorflow, Keras, RTMaps
Strengths	Image analysis, mathematical morphology, signal processing, graph theory, machine learning, deep learning.
Languages	English(Professional), French(Fluent, TCF-ANF Certified), Tamil(Maternal), Kannada, Hindi(Fluent).
Domains	Video representation learning, time series analysis, hyperspectral imaging, Graph cuts, image segmentation.
Reviewer	CIARP 2012, ISMM 2013/15, PR Letters 2013/15/16, ICIP 2014, ITSC 2016, MSSP 2017, ICVES 2017, IROS 2017, TIP 2017, NIPS2017 LLD,ITSC 2018, PR 2018, ALA 2018.
Teaching	Machine learning, Dimensionality reduction course, Image processing course

Experience

Jan 2018 – Now **AI-ML R&D Engineer, AKKA Technologies**, *Paris area, France*.
- Overseeing & guiding team's integration of S/W components for autonomonous driving systems. [Project]
- Co-defined & prototyped 3D prior maps to localize static obstacles for autonomous driving.
- Co-defined & prototyped clustering and obstacle detection pipeline and tracking with Lidar.
- Reviewed Deep Reinforcement Learning (DRL) & End-to-End DNN architectures for Autonomous Driving Tasks.
- Co-encadrer des doctorats et des stages de master tout en visant la recherche industrielle.

May 2017 – Nov 2017 **Consultant (Remote) at Uncanny Vision**, *Paris, France*, Project.
- **Problem** : Video representation learning for unsupervised anomaly detection in video surveillance context.
- **Models** : Variational Autoencoders, Generative Adversarial Networks, Convolutional LSTM
- **Collaboration** : Ranjith Parakkal, Dilip Thomas

Oct 2016 – Aug 2017 **Teaching assistant & Researcher(ATER), Université Lille 3**, *Lille, France*.
- **Teaching** (192H) : (License/Master) Data mining, Machine learning, Dimensionality reduction.
- **Lab** : Data intelligence Lab, CRIStAL, Lille **Focus** : Time series analysis and sequential learning.
- **RF pruning** : Evaluated performance of pruned random forests using their out-of-bag samples. [git] [Project]

Dec 2015 – Nov 2016 **Post-doc Data lab, ENS Paris and ThalesAlenia Space**, *Paris, France*.

- **Problem** : Time series anomaly detection.
- **Streaming Multiscale Anomaly detection** [Project]: Track correlations across multiple scales of a moving window over time series with varying pseudo-periodicities. A streaming PCA algorithm to decorrelate the reconstruction errors across scales to detect anomalies. [**Datasets** Yahoo! and Numenta]
- **Predictive models for anomaly detection(AD)** : **Data** : Telemetry data from ThalesAlenia Space.
  - Auto-regressive models on principal components using scattering transform, Non-linear moving window features.
  - Other tasks : data cleaning, visualization, validating results with domain experts.
- **Deep learning reading group** reviewed state-of-the-art in CNN, RNN architectures & generative models.
- **Deliverables** : Predictive models, Streaming AD, Scattering transform, Local Outlier Factor, final report.

Nov 2014 – Nov 2015 **Post-doc Centre de Robotique(CAOR) & HELICoiD, Mines ParisTech**, *Paris, France*.

- **Project** : Real-time classification of hyper-spectral images(HSI) for surgical inter-operative aid and Tumor detection.
- **Advisors** : Bogdan Stanciulescu, Jesus Angulo
- **Description** : The goal of HELICoiD project is to localize cancerous tissues using high spatial-spectral resolution HSI of the brain in-vivo, for better tumor resection margins. Hierarchical NMF was used as an exploratory tool to extract the best rank-1 approximation of the input HSI pixel subsets.
- **Deliverables** : Unsupervised clustering by H2NMF, Random Forest cluster classifier pipeline, Data processing : spatio-spectral repeatability error estimation, camera calibration, specularity removal. Final project report.
- Worked across multi-disciplinary team of neurosurgeons imaging, algorithms & architecture experts.

Oct 2011 - Oct 2014 **PhD at A3SI-LIGM UMR 8049, ESIEE**, *Paris,France*.

- **Title** : *Energetic-Lattice based optimization* [Thesis] [Slides] **Director** J. Serra **Co-encadrant** J. Cousty
- **Teaching** (100H) : TD, TP and Masters Courses in Mathematical Morphology & Graph Theory.
- **Contributions** : Generalized Breiman's Dynamic program (DP) for image image segmentation by characterizing energies and partition families that can be optimized by the dynamic program. Evaluation on Berkeley's dataset and geo-spatial population data from PACA.
- **Adapting hierarchy with multi-label graph cuts** : Adapting hierarchy of segmentations with hierarchical costs (parent-child relationship) using multi-label graph cuts.

Apr – Jun 2011 **Internship at A3SI-LIGM, ESIEE**, *Paris, France*.

Tutorial on Morphological operators in PINK, a library for image processing operators and non-linear filtering.

Mar 2010 – Feb 2011 **Research Assistant, Computer Vision and AI Lab, IISc**, *Bangalore, India*.

- **Project** : Segmentation algorithms for roads/lanes for autonomous cars. **Advisor** : K. R. Ramakrishnan
- **Industrial project**: Compressed domain H.264 motion detection for video surveillance.
- **Mentored** : 3 bachelors and 1 masters project on the two problems.

Aug 2008 – Feb 2010 **Software Design Engineer, Texas Instruments**, *Bangalore, India*.

Embedded systems and programming on ARM and DSP.

Education

2011–2014 **PhD in Computer Science, Université Paris-Est Marne-la-Vallée**, *Paris, France*.

Best thesis award from Université Paris-Est 2015 in Mathematics et STIC

2004–2008 **B.E. Electronics & Communication, Visvesvaraya Technological University**, *Bangalore, India*.

- Grade: 3.8/5, Passed with distinction, Best outgoing student.
- Thesis: Crosstalk Elimination in ADSL systems by wavelet packet techniques.

Publications

Journals

1. Overview of deep learning based methods for unsupervised and semi-supervised anomaly detection in videos, B Ravi Kiran, Dilip Thomas, Ranjith Parakkal, Journal Imaging MDPI Jan 2018, [pdf].
2. Spatio-spectral classification of hyperspectral images for brain cancer detection, PLOS one [pdf] March 2018
3. Intra-operative Visualization Using HSI for Brain Tumor Delineation, H. Fabelo, et al. [pdf], Sensors, MDPI, Jan 2018.
4. Global-local optimizations by hierarchical cuts, B Ravi Kiran, J. Serra , Pattern Recognition, Jan 2014 [Link]
5. Fusion of ground truths & hierarchies, B Ravi Kiran, J. Serra, Pattern Recognition Letters, Oct 2014 [Link]

Conferences

1. Streaming multi-scale anomaly detection for univariate time series, B Ravi Kiran, CAp 2017 [pdf, slides, poster]
2. Cost-complexity pruning of Random Forests, ISMM 2017, B Ravi Kiran, J. Serra.
3. Spatio-Spectral Classification of HSI [1] by Supervised & Unsupervised Methods. DCIS 2016, S. Ortega, et al. (3rd author)
4. Brain Cancer Detection based on Spatial-Spectral HSI Classification, DCIS 2016 Fabelo H., et al. (4th author)

[1] Hyperspectral Images

5. Digitization of partitions & tessellations, Jean Serra & B Ravi Kiran, DGCI 2016 [pdf]
6. Clustering of HSI of brain tissues by hierarchical NMF, B R Kiran, B. Stanciulescu, J. Angulo, BIOIMAGING 2016,[pdf]
7. Braids of partitions, B Ravi Kiran, J. Serra, ISMM 2015 [pdf].
8. Constrained optimization on hierarchies of partitions, J. Serra, B Ravi Kiran, ISMM 2015, [pdf]
9. Energetic lattice for optimizing over hierarchies of partitions: J. Serra, B Ravi Kiran, ICIP 2014 [link]
10. Scale Space Operators on hierarchies of segmentations, B Ravi Kiran, Jean Serra, SSVM 2013, [pdf] [Poster]
11. Ground truth energies for hierarchies of segmentations, B. Ravi Kiran, Jean Serra, ISMM 2013. [pdf] [Poster]
12. Optima on hierarchies of partitions, J. Serra and B. Ravi Kiran, ISMM 2013. [pdf]
13. Global constraints on hierarchical segmentation, B Ravi Kiran, J. Serra, J. Cousty, ECCV 2012, HiPOT WK. [pdf]
14. Hierarchies & climbing energies, J. Serra, B Ravi Kiran ,J. Cousty, CIARP 2012 [pdf]
15. Summarizing Cricket Videos, Y S Kumar, S K. Gupta, **B R Kiran**, K R Ramakrishnan, C. Bhattacharyya, ISCE 2011 [Link]
16. Parallelizing connectivity Operators for Multicore Envs, Anoop K. P., **B R Kiran** & Y. Senthil Kumar, ICCSP 2011. [Link]
17. Connected Component Labeling by Recursion, B Ravi Kiran, Y S Kumar, Anoop K P, K R Ramakrishnan, NCC 2011 [Link]

Invited Talks

- Streaming multi-scale anomaly detection on time series, [Slides], CRISTaL Lille UMR 9189 May 2017
- Hierarchical clustering of hyperspectral images for tumor detection [Slides], Icube Univ. Strasbourg Mar 2017
- Braids of partitions and applications : Indian Institute of Science, EE Department, Aug 2016
- Constrained Optimization on Hierarchies of partitions [slides] : Centre de Morphologie Mathematique, Jan 2015.
- Fusions of Ground Truths and of Hierarchies [slides] : Journée ISS France, Ecole des Mines de Paris, Feb 2014.
- Climbing energies and optimal cuts [Slides 1, Slides 2]: UPC, Barcelona, Image Processing Group, Jun 2013.
- Climbing energies and optimal cuts : CMM, Fontainebleau, March 2013.
- Ground truth energies for hierarchies of partitions : Journée ISS France, Mines ParisTech, Paris, Feb 2012.
- Optimization on hierarchies & GIS problems : Indian Institute of Science CVAI lab Bangalore, India, June 2012.

Tutorials, Posters, Reports

- Tutorial ICIP 2014, Paris, *Optimizations on Hierarchies*, B. Ravi Kiran, J. Serra, J. Cousty, & H. Talbot. [slides]
- Theory of Braids, Energetic Lattices & Constrained Optimization, J. Serra, B Ravi Kiran. [Part 1] [Part 2]
- End-member extraction in HS images for tumor detection Kiran B.R., Stanciulescu B., Angulo J. ROMOPTO 2015 [poster]
- Real-Time Background Subtraction Using Adaptive Sampling & Cascade of Gaussians, B Ravi Kiran, Y Senthil Kumar [pdf]
- Climbing on Pyramids, J. Serra, Bangalore Ravi Kiran, Technical Report. 2012 [pdf]

References

- Industry : Senthil Yogamani, Technical Lead, Computer Vision Architect for Autonomous Driving, Valeo, Galway, Ireland. [LinkedIn]
- Industry : Laurent Guigues, Principal research scientist, Amazon Seattle, United States. +1 408 647 0724, [LinkedIn]
- Post-doc Advisor : Jesus ANGULO, Senior scientist, Centre de Morphologie Mathematique, MINES ParisTech. + 33 1 64 69 47 75
- PhD advisor : J. SERRA, Professor Emiritus, Informatique, ESIEE Université Paris-Est. +33 (01) 6423 4820
- PhD reporter : Philippe SALEMBIER, Professor, Universitat Politecnica de Catalunya, Barcelona +(34) 9 3405 40 30

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